SLA Printing

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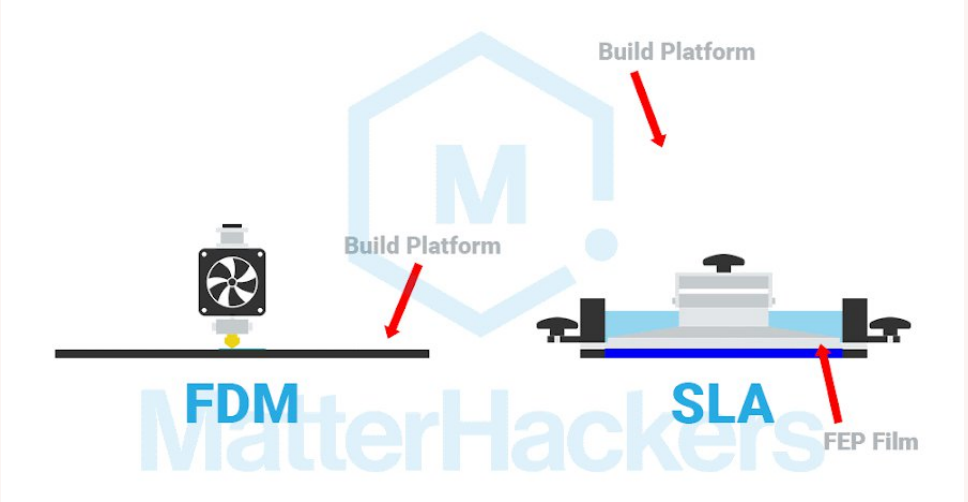
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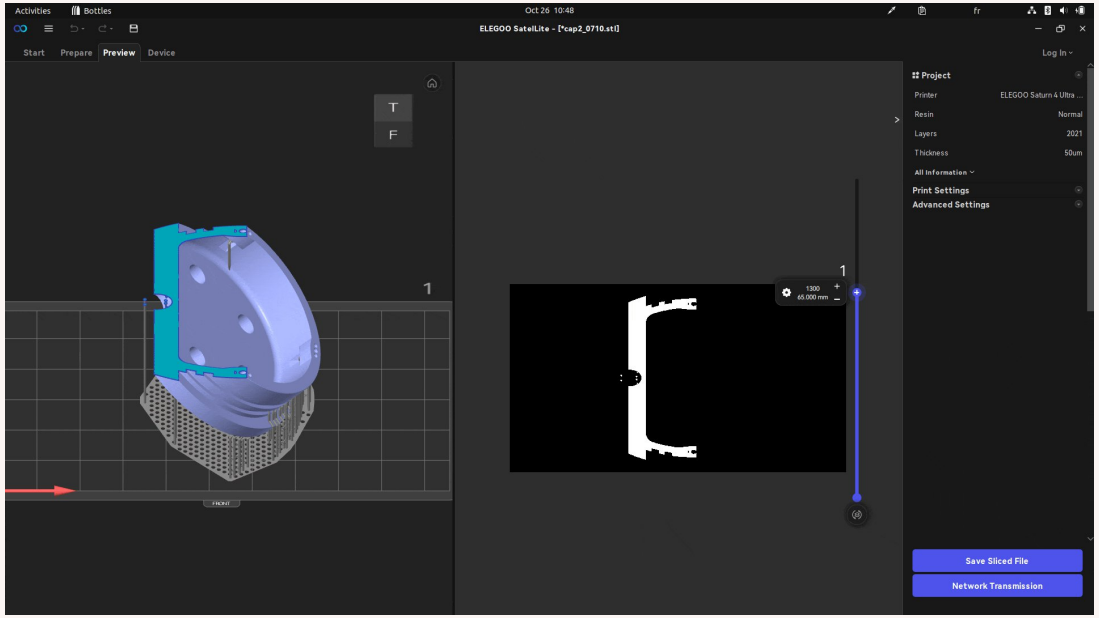
- **What is SLA 3D printing ?**
- **When is it relevant to use ?**
- **How to use the SLA printer ?**
  - Safety measures
  - Software
  - Processes
  - Maintenance



What is SLA 3D printing ?

- SLA printers use liquid UV resin
- The build platform is facing down on the screen
- An UV LCD screen display a black and white image corresponding to the layer
- Each layer is exposed a certain amount of time to let the resin cure
- The build platform raises until the model is printed





When is it relevant to use ?

- **Advantages :**
  - Great resolution
  - Homogeneous solid (no layer weakness, watertight)
  - Various materials to choose from

- **Drawbacks :**
  - Complex handling and post processing
  - Safety issues
  - Hard to get precise dimensions

- **When is it useful for robots ?**
  - For watertight or smooth parts
  - If a specific material is needed



How to use the SLA printer ? - Safety measures

- **Health Hazard/Hazardous to the Atmospheric System**
  - Acute toxicity (oral, dermal, inhalation)
  - Skin irritation
  - Eye irritation
  - Skin sensitization
  - Respiratory tract irritation
  - Narcotic effects
- **Hazardous to the Environment**
  - Acute hazards to the aquatic environment
  - Chronic hazards to the aquatic environment
- **Flammable**









How to use the SLA printer ? - Safety measures

- **Mandatory safety measures :**
  - Butyle gloves
  - Safety glasses
  - Respiratory mask
  - Lab coat
  - Ventilation

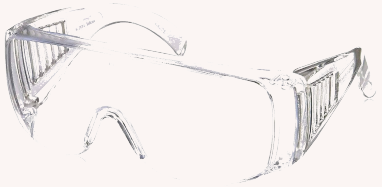




- **Consider the workspace as contaminated**

- **Every resin waste belong in the hazard bin**

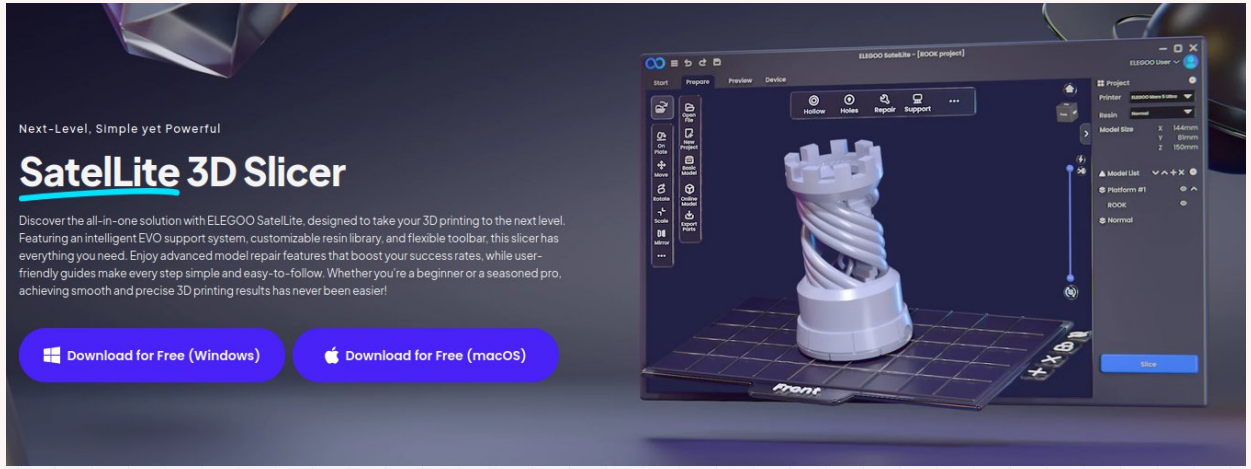


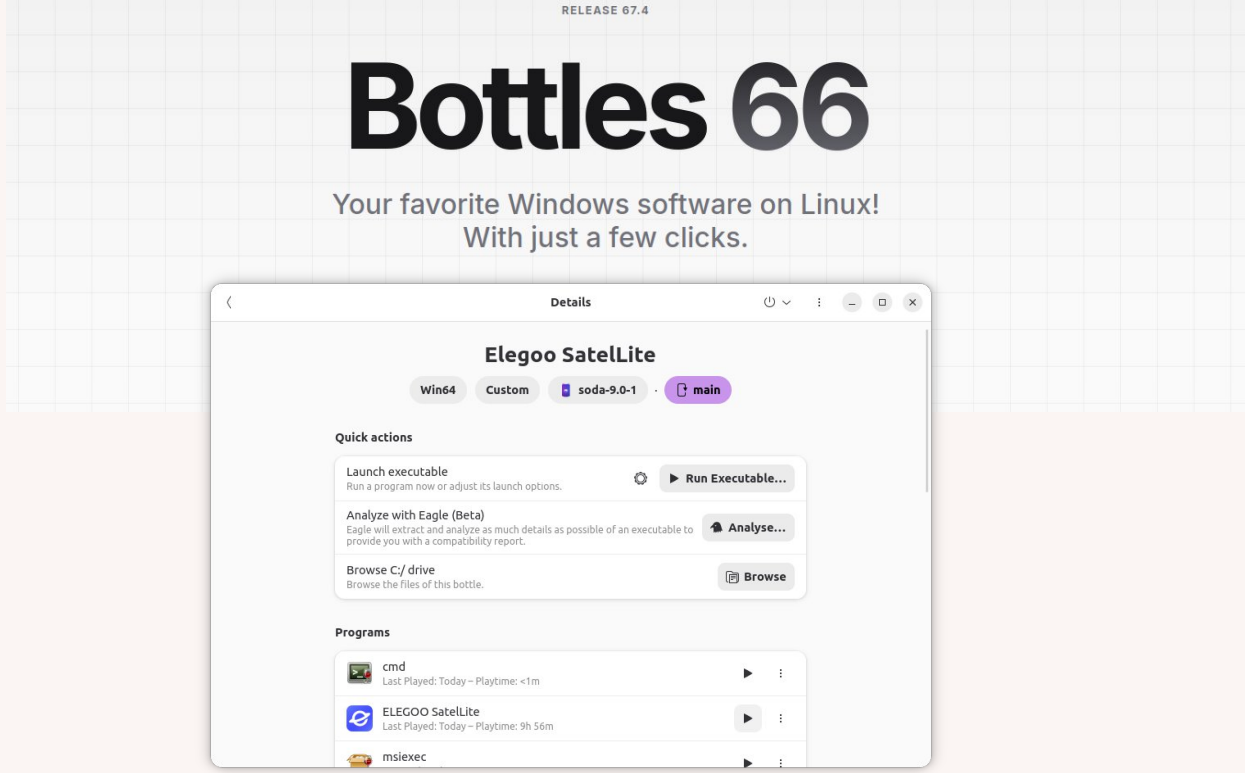


- **No liquid that has resin should be disposed of in the sink**

How to use the SLA printer ? - Software

- **Elegoo SatelLite 3D Slicer**
  - Native for Windows
  - For Linux, use with Bottles (soda-9.0-1)

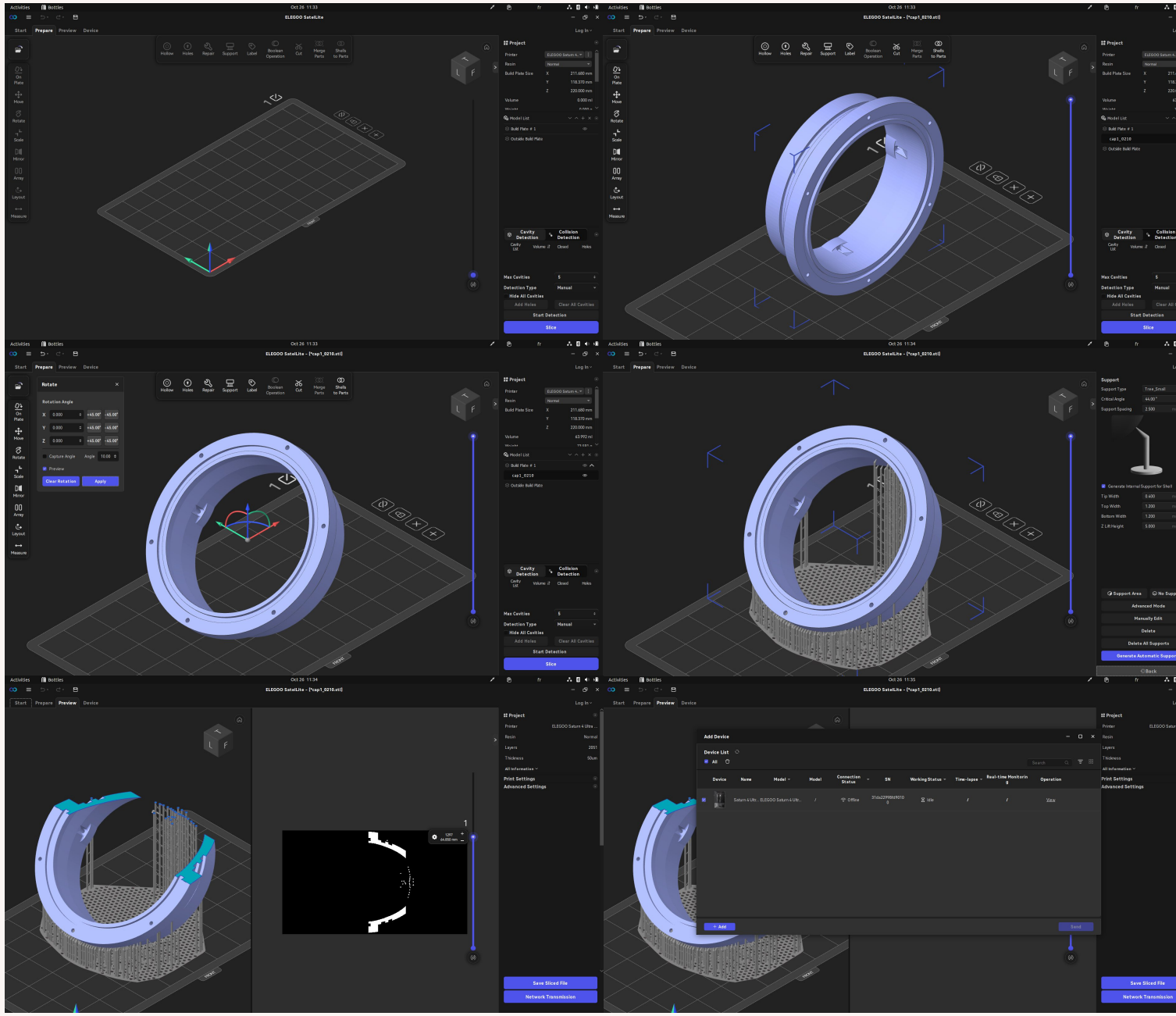




How to use the SLA printer ? - Software

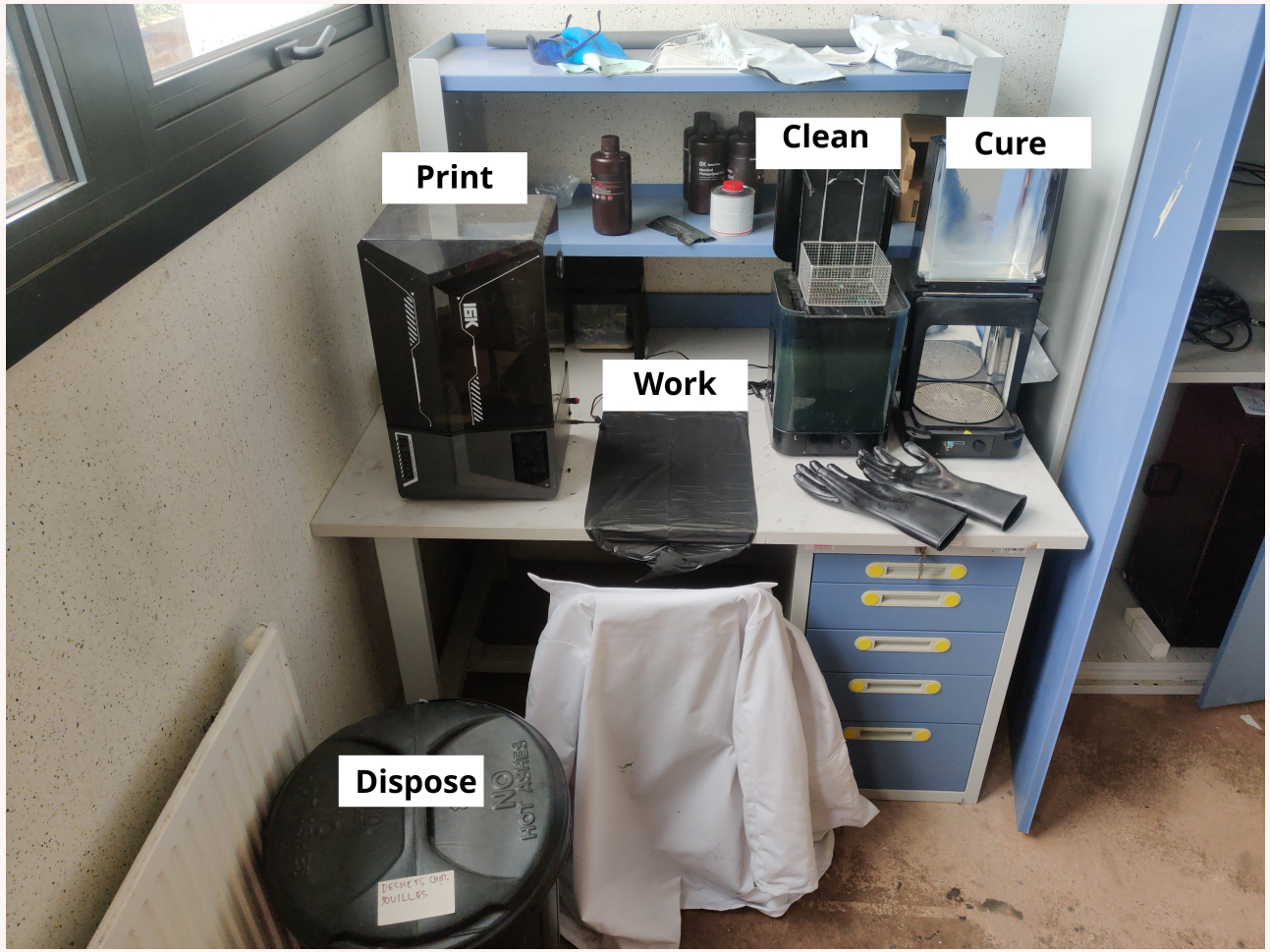
- **Elegoo SatelLite 3D Slicer**
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- **Software workflow :**
  - Export desired part in .STL
  - Import in SatelLite
  - Rotate to guarantee correct printing
  - Generate support
  - Slice
  - Send via network to the printer



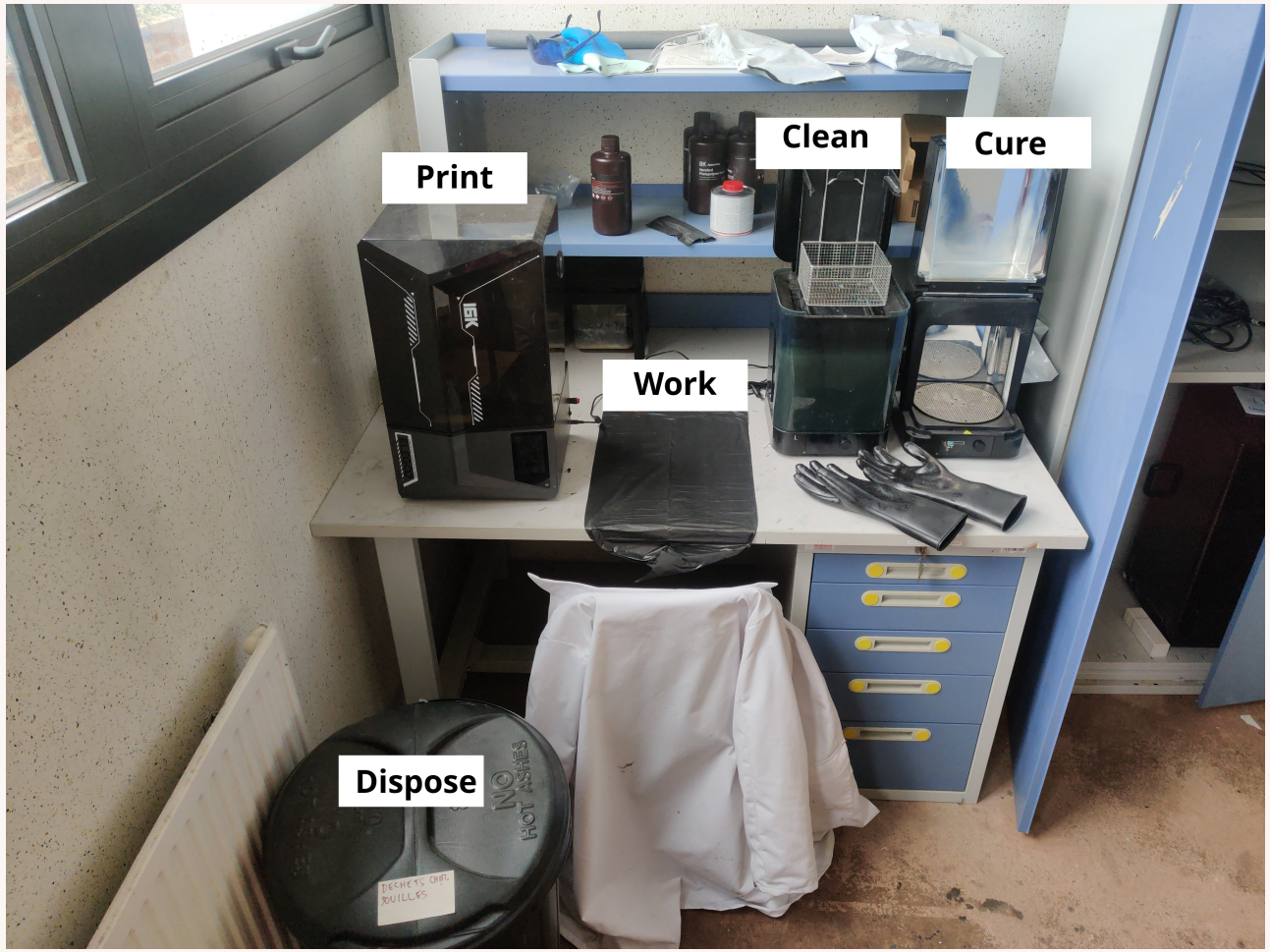
How to use the SLA printer ? - Processes

- **Print**
  - Wear all safety equipments
  - Check that the resin level is between MIN and MAX
  - Check that the film is clean
  - Select sent file and print
- **Clean**
  - Wear all safety equipments
  - Prepare the workplace with a disposable trash bag
  - Prepare paper towel with isopropylic alcool
  - Remove the build plate
  - Remove the printed part on the trash bag
  - Clean the build plate and put it back on the printer
  - Remove supports
  - Wash the parts in the washer
- **Cure**
  - Wear all safety equipments
  - Once cleaned, wipe dry the parts
  - Cure the parts in the curer



How to use the SLA printer ? - Maintenance

- **Maintenance**
  - The printer film is delicate, it should be replaced frequently (layer count on the printer)
  - The clean tank should be replaced (WIP)
  - The waste should be emptied (WIP)



Thank you for your attention

Do you have any question?

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